ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 2 · PRACTICE E

MARKS: 80 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET second paper (practical, 80 marks): breadboard prototypes of a 555 alternating LED flasher, a half-wave power supply with a Zener regulator, a three-LED battery-level indicator, and a DPDT motor-reversing circuit with its circuit diagram.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.
2. This is a practical paper. Use Fritzing for breadboard prototypes and draw.io (or similar) for circuit diagrams. If you do not have a computer, draw the breadboard layouts and circuit diagrams neatly by hand.
3. Put your name or examination number at the top of every printout or page.
4. Show the positive and negative supply rails clearly, place every component correctly and make all the connections.
5. Save your work regularly. In Fritzing, the AC supply may be shown as two labelled wires.

QUESTION 1

The circuit below uses a 555 timer in astable mode to flash two LEDs one after the other. Design and print its breadboard prototype.

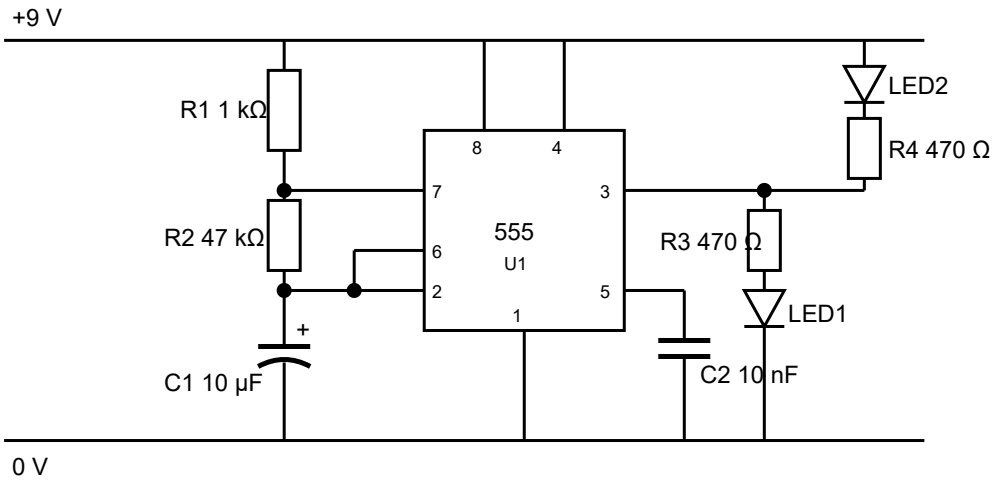


FIGURE 1: 555 alternating LED flasher

Components: 1 × 9 V battery; 1 × NE555 timer IC; 1 × 1 kΩ (R1), 1 × 47 kΩ (R2) and 2 × 470 Ω (R3, R4) resistors; 1 × 10 µF electrolytic capacitor (C1); 1 × 10 nF capacitor (C2); 2 × LEDs; breadboard and jumper wires.

Design and print (or draw) the breadboard prototype of the circuit in FIGURE 1. (20)

[20]

QUESTION 2

The circuit below is a half-wave rectifier with a smoothing capacitor and a Zener regulator that gives about 5,1 V to an LED load. Design and print its breadboard prototype.

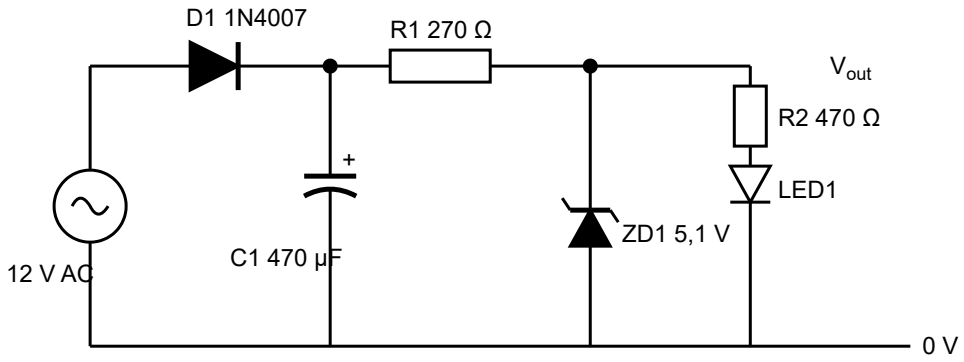


FIGURE 2: Half-wave rectifier with Zener regulator

Components: 12 V AC supply (two labelled wires); 1 × 1N4007 diode; 1 × 470 µF electrolytic capacitor; 1 × 270 Ω resistor (R1); 1 × 5,1 V Zener diode; 1 × 470 Ω resistor (R2); 1 × LED; breadboard and jumper wires.

Design and print (or draw) the breadboard prototype of the circuit in FIGURE 2. (20)

[20]

QUESTION 3

Nomvula wants an indicator that shows how much charge is left in her robot's 9 V battery. She uses three branches across the battery. Each branch has a Zener diode (cathode towards +9 V), a 470 Ω resistor and an LED in series. The green branch has a 6,2 V Zener, the yellow branch a 4,7 V Zener and the red branch a 3,3 V Zener. As the battery runs down, first the green LED goes off, then the yellow, and finally the red.

Components: 1 × 9 V battery; 3 × Zener diodes (3,3 V, 4,7 V and 6,2 V); 3 × 470 Ω resistors; 1 × red, 1 × yellow and 1 × green LED; breadboard and jumper wires.

Use Fritzing (or draw by hand) to design the breadboard prototype of Nomvula's battery-level indicator. Add your name or examination number and print. (20)

[20]

QUESTION 4

A small robot conveyor must be able to run forwards and backwards. Thabo uses a 6 V battery pack, a 1 A fuse, an ON/OFF switch, a DPDT (double-pole, double-throw) centre-off switch wired as a reversing switch, and a 6 V DC motor. A red LED with a 330 Ω resistor shows that the power is on.

Components: 1 × 6 V battery pack; 1 × 1 A fuse with holder; 1 × SPST ON/OFF switch; 1 × DPDT centre-off switch; 1 × 6 V DC motor; 1 × red LED; 1 × 330 Ω resistor; breadboard and jumper wires.

4.1 Draw the circuit diagram of the reversing circuit (in draw.io or neatly by hand), using the correct symbols. (5)

4.2 Design the breadboard prototype of the reversing circuit in Fritzing (or draw it by hand). (15)
Add your name or examination number and print.

[20]

TOTAL: 80

END OF QUESTION PAPER